SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – INDIVIDUAL TALK

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 3 – INDIVIDUAL TALK
Weightage:	30 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	IT-2024	**Date:**	

Code of Conduct

*I **Sri Janani B (192421036)** (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the Student: **Sri Janani B***

Instructions

- Topic given before the session and students speak on stage.
- Duration: 10 minutes (In class)

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Security and Authentication Mechanisms in Cellular Networks



SYSTEM VIEW

Where Security Lives in the Network

Subscriber side

User equipment

The USIM stores the secret key K and the subscriber identity.

Radio side

Access network

The base station (eNB or gNB) protects the radio link.

Operator side

Core network

AMF, AUSF and UDM authenticate users and manage keys. In 4G these are the MME and HSS.

SECURITY GOALS

Confidentiality | Integrity | Mutual authentication | Subscriber privacy

EVOLUTION

Security from 2G to 5G

Generation	Authentication	Encryption	Known weakness
2G GSM	One-way: the SIM proves itself to the network	A5/1 and A5/2 stream ciphers	No network authentication, weak ciphers
3G UMTS	Mutual authentication with AKA	KASUMI block cipher	Falls back to 2G, IMSI exposure
4G LTE	EPS-AKA with a key hierarchy	AES, SNOW 3G, ZUC	IMSI sent in clear at attach
5G NR	5G-AKA and EAP-AKA′	AES, SNOW 3G, ZUC	Interworking with older networks

4G LTE

EPS-AKA: How an LTE Phone Is Authenticated

1 Attach request
The phone sends its identity (IMSI or GUTI). The MME asks the HSS for an authentication vector.

2 Vector generated
The HSS creates RAND, AUTN, XRES and the key KASME from the secret key K.

3 Network challenge
The MME sends RAND and AUTN. The USIM checks AUTN, which proves the network is genuine.

4 Response and keys
The phone returns RES. If it matches XRES, both sides derive session keys from KASME.

> The secret key K never leaves the USIM and the operator's authentication center.

5G NR

5G Authentication: Stronger and More Private

5G-AKA
An evolution of LTE AKA. The home network confirms the authentication result to the serving network.

EAP-AKA'
An extensible option that lets non-3GPP access, such as Wi-Fi, reuse SIM credentials.

SUCI concealment
The permanent ID (SUPI) is encrypted with the home network's public key, so the IMSI is no longer sent in clear.

KEY HIERARCHY



PROTECTING TRAFFIC

Encryption and Integrity Protection

ALGORITHM SETS (5G NAMES)

Algorithm	Ciphering	Integrity
SNOW 3G	128-NEA1	128-NIA1
AES	128-NEA2	128-NIA2
ZUC	128-NEA3	128-NIA3

NEA means encryption and NIA means integrity. 4G uses the equivalent EEA and EIA names.

Signaling is protected

Control-plane messages (NAS and RRC) are both encrypted and integrity protected.

User data is encrypted

User-plane encryption is standard. User-plane integrity protection is new in 5G and optional.

Ready for the future

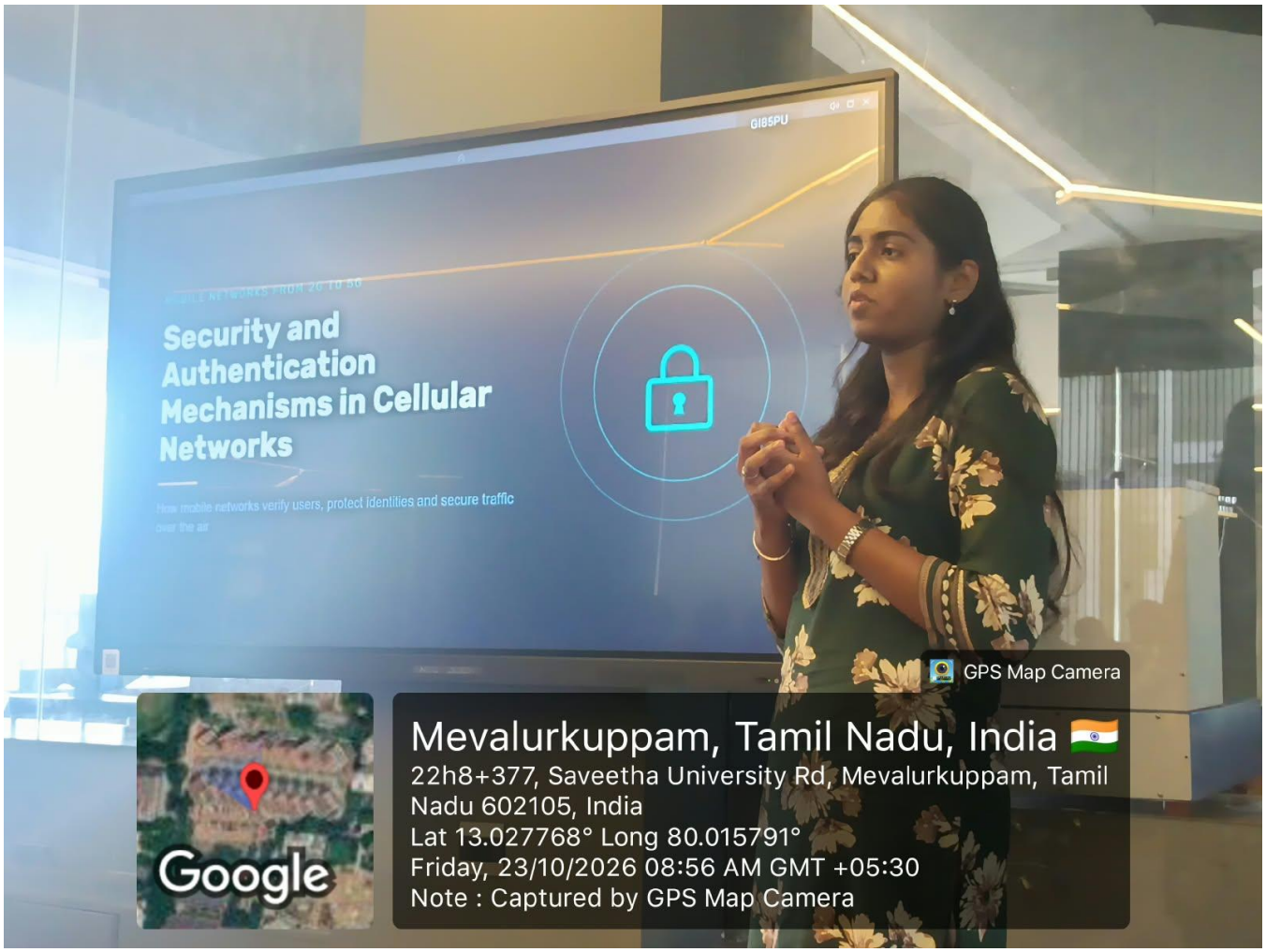
The 5G design allows 256-bit keys, anticipating stronger algorithms.

ATTACKS AND DEFENSES

Known Attacks and Countermeasures

Threat	Countermeasure
IMSI catching with rogue base stations	SUCI conceals the permanent identity in 5G
Man-in-the-middle on the radio link	Mutual authentication and integrity protection
Signaling abuse over SS7 and Diameter	Signaling firewalls; SEPP secures the 5G interconnect
Downgrade to 2G	Disable legacy 2G where possible and keep devices updated

Takeaway: each generation closed earlier gaps, and today's risks sit mostly in legacy interworking and deployment.


GIBSPU
Security and Authentication Mechanisms in Cellular Networks
How mobile networks verify users, protect identities and secure traffic over the air
GPS Map Camera
Google
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